Aniketh Bandlamudi

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Education

University of Virginia Charlottesville, VA

B.S. in Computer Science & Applied Mathematics, School of Engineering and Applied Science *Expected 2029*

- *Relevant Coursework:* APMA 2120 (Multivariable Calculus), CS 2100 (Data Structures & Algorithms 1)
- *Activities:* UVA Solar Car Team (Embedded Software), IEEE UVA Student Chapter (Member)

Experience

Founder & Engineer Jan. 2026 – Present

Pragmatic (aniketh.net/pragmatic) *Remote*

- Building a zero-dependency durable execution runtime in Rust that lets nondeterministic AI agents survive crashes and resume exactly where they left off
- Designed an append-only step journal at 230 ns/append with flat latency through 1M entries, passing a 1000/1000 replay-determinism test suite
- Formalized agent execution and replay with a probabilistic process calculus
- Incorporated as Pragmatic Research, Inc. (Delaware C-corp) in Aug. 2026

Engineering Intern May 2026

Alarm.com *Tysons, VA*

- Configured and integrated home security devices for hardware QA testing
- Soldered buck converters and embedded boards in the hardware lab
- Built an internal software tool to manage lab devices, streamlining QA/QE-to-device-engineering testing cycles

Founder & President May 2022 – Present

OCRadar (ocradar.com) *Vienna, VA*

- Founded a nonprofit and led a team of 4 building a computer-vision mobile platform for early oral cancer detection
- Published research on a CNN that detects oral squamous cell carcinoma with 98% accuracy, paired with custom hardware for consistent image capture (arXiv:2510.16235)
- Featured in outlets reaching millions: WTOP, Korea Times, Vision Times China, FCPS News, Tech Roast Show

Software Lead (FRC) / R&D Captain (FTC) 2022 – 2026

Oakton Robotics *Vienna, VA*

- Led software for a 25-member team, owning systems engineering and control software for competition robots
- Oversaw robot design and programming over four seasons; earned finalist, innovation, and creativity awards
- Ran STEM workshops and community outreach programs

Projects

Haze | *Swift, iOS, watchOS* Site | App Store

- Solo-built and shipped a 5-star iPhone and Apple Watch weather app with widgets, Live Activities, and Siri
- Blends ECMWF, GFS, and ICON models, corrects with NWS station data, and sends background rain alerts

Meridian | *Rust, Swift, UniFFI, DuckDB, Claude API*

- Built a macOS financial terminal on a 53K-line Rust core bridged via UniFFI to SwiftUI, with 35+ functions
- Streams 2,000 live symbols at 60 fps on under 10% of one CPU core from 9 data sources, with a Claude AI analyst

Hypnagogia | *Python, Brian2, TypeScript* Live | GitHub

- Simulated a whole Drosophila brain (144K neurons, 21.9M synapses) to test if a learned memory replays in sleep
- Scored replay against shuffled-connectome and random-ensemble controls; reported negative results in full

IllumiSign | *Hack the Nest 2025 Winner* Devpost

- Won Hack the Nest 2025 with a browser-based communication tool that uses real-time eye tracking to let ALS patients select letters and generate speech without specialized hardware

Technical Skills

Languages: Java, C++, Python, Rust, Swift, TypeScript, HTML/CSS/JavaScript
Technologies: PyTorch, Computer Vision, iOS/watchOS Development
Robotics & Hardware: WPILib, PathPlanner, Limelight, CTRE Phoenix, SysId, Embedded Systems, KiCad
Developer Tools: Git, Xcode, VS Code, Claude Code, Codex, Docker, ffmpeg, COLMAP
Design & CAD: Figma, Pixelmator Pro, AutoCAD, Autodesk Inventor, Fusion 360